Assignment Video No- 41) Hands-On Decision Tree

Input Dataset = 50_Startups.csv

1. **Multiple Linear Regression** – R Score Value = **0.9358680970046243**
2. **SVM (Support Vector Machine)**

S.No	Hyper Parameter	Linear (R Score value)	rbf - Non Linear (R Score value)	Poly (R Score value)	Sigmoid (R Score value)
1	Nothing(I didn't use 'C' parameter)	0.895077924	-0.057418394	-0.057103875	-0.057209359
2	**C=0.10**	**0.937521652**	-0.0574795	-0.057448044	-0.057458595
3	C=10	-2.437215013	-0.056807593	-0.019802139	-0.054719583
4	C=100	-357.0795116	-0.050726023	0.114684807	-0.030453515
5	C=500	-8996.86065	-0.024323348	0.266163709	0.070572145

3. **Decision Tree**

S.No	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1	MSE (squared_error)	None (Auto)	random	0.789717484
2	MSE (squared_error)	sqrt	random	-0.853959879
3	MSE (squared_error)	log2	random	0.244993297
4	**MSE (squared_error)**	**None (Auto)**	**best**	**0.903740762**
5	MSE (squared_error)	sqrt	best	0.34898833
6	MSE (squared_error)	log2	best	0.449400769
7	MAE (absolute_error)	None (Auto)	random	0.93516123
8	MAE (absolute_error)	sqrt	random	0.749488238
9	MAE (absolute_error)	log2	random	0.067197666
10	**MAE (absolute_error)**	**None (Auto)**	**best**	**0.951232654**
11	MAE (absolute_error)	sqrt	best	0.680644498
12	MAE (absolute_error)	log2	best	0.568286283
13	variant of MSE (friedman_mse)	None (Auto)	random	0.893237586
14	variant of MSE (friedman_mse)	sqrt	random	0.081767937
15	variant of MSE (friedman_mse)	log2	random	0.690716789
16	variant of MSE (friedman_mse)	None (Auto)	best	0.903476105
17	**variant of MSE (friedman_mse)**	**sqrt**	**best**	**0.947249009**
18	variant of MSE (friedman_mse)	log2	best	0.827195651
19	poisson	None (Auto)	random	0.527439268
20	**poisson**	**sqrt**	**random**	**0.959709603**
21	poisson	log2	random	-1.076243541
22	poisson	None (Auto)	best	0.932159387
23	poisson	sqrt	best	0.584696093
24	poisson	log2	best	0.551134866

“Finally, we found the correct model is ‘Decision Tree’ for this dataset, where the parameters are: criterion = poisson, max_features = sqrt, and splitter = random.”